

Static and Dynamic Calibration of 9-Axis IMU using UKF with Low Cost Setup

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Abstract— MEMS based IMU sensors are very popular for its small size, low-cost and low power consumption. However, for accurate and precise data collection, proper calibration of IMU is compulsory. In this paper we have proposed a technique for both static and dynamic calibration of IMU sensor. We have developed a simple low-cost setup and used least square (LS) fit and ellipsoidal fit algorithm for static calibration and further incorporated Unscented Kalman filter (UKF) for dynamic calibration. The accelerometer is calibrated using 108 position data for three axis and gyroscope is calibrated using specially designed low-cost setup with LS algorithm whereas for magnetometer calibration, ellipsoidal fitting algorithm is used. Accordingly, all the calibration parameters like bias factor, scale factor and misalignment for accelerometer; bias factor, scale factor and non-orthogonality for gyroscope; Offset, gain and cross-gain for magnetometer are obtained. Moreover, for dynamic models like real-time or sensor fusion applications, the UKF (Unscented Kalman filter) based algorithm is proposed to obtain the calibration parameters dynamically.

Keywords— IMU, calibration, scale factor, bias factor, least square, ellipsoid fit, UKF.

I. INTRODUCTION

The Inertial Measurement Unit (IMU) is a crucial component in navigation systems and medical appliances. The Inertial Measurement Unit (IMU) comprises of three-axis accelerometer, three-axis gyroscope making it as a 6-axis IMU and some may include three-axis magnetometer making it as a 9-axis IMU. Although MEMS based IMU is very popular because of its small size, low-cost and low power consumption, the major issues of IMU are its prone to drift and various error factors which typically includes Bias, Scale factor and Misalignment error. To overcome the limitation and acquire precise data, we need to determine the parameters like scale factor, bias factor, misalignment error, offset, gain, cross-gain accurately and put it in sensor error model to get the calibrated data. An uncalibrated data refers to the raw measured data obtained by a sensor that contains the true physical quantity together with undesired inaccuracies whereas the calibrated data represents the error free data. There are different calibration methods for IMU in the literature which are discussed as follows. Cheuk et. al., conducted a study on automatic calibration for inertial measurement units [1]. In that study, the sensor was placed in 12 different orientations to determine the scale factor,

bias factor, and misalignment. Skog et. al., presented calibration of a MEMS inertial measurement unit [2] but here for calibrating the gyroscope the setup is rotated manually which may not be convenient to have precise speed control and thus it may induce error. Similarly, to calibrate the gyroscope Qureshi et. al., [3] used Single axis rate table by ideal Aerosmith but it is very costly. Tsai C L et al. proposed an adaptive and learning calibration method for magnetometers using ellipse fitting. Merayo et al. proposed a magnetometer calibration method that utilizes least square to estimate the ellipsoid parameters. Wu et al suggested magnetometer calibration using least square method, which considers the presence of measurement errors.

In [1], the authors used only 12 position data for calibrating. To improve the accuracy further in predicting the calibration parameters, larger number of positions and data can be obtained. In [2], the researchers used manual rotation of set-up which may lead to sudden jerk-off or change in reference velocity. In [3], very costly equipment was used for calibration. In [4], the authors used ellipse fitting so complete correction of error parameters is not possible due to the utilization of only two-dimensional curve. In [5] and [6], the authors used Least square method, which introduces the presence of the error parameters. In-order to overcome this issue an optimal ellipsoidal fitting approach that employs a condition constraint. The magnetic interference parameters are derived by limiting the three-dimensional surface to an ellipsoid using the least square method. Further most of the existing literature focus only on the static calibration.

To overcome the above limitation, we propose a low-cost setup and least square, ellipsoidal fitting-based algorithm along with UKF for both static and dynamic calibration. The system includes a motor and a unique drive mechanism that allow us to accurately calibrate the gyroscope without experiencing any jerks. We are keeping track of 108 different positions when calibrating the accelerometer in order to obtain the scale factor, bias factor, and misalignment for increased accuracy. The developed set-up is very cost effective, user-friendly and portable. Further for navigation and medical application where only static calibration is not sufficient, we propose UKF based

algorithm to estimate the sensor biases, scale factors, and misalignments dynamically and precisely.

The paper is organized as follows: Section II presents calibration algorithm, Section III discusses on experimental setup and calibration procedure, Section IV presents the experimental results followed by conclusion in Section V.

II. CALIBRATION ALGORITHM

A. Static Calibration

For Accelerometer and Gyroscope Least square fit algorithm [9] – [10] is followed whereas for magnetometer Ellipsoidal fit [11] algorithm is adapted.

1) Accelerometer calibration

The Error Model of a Tri-axial accelerometer can be represented by

$$\begin{bmatrix} Ax_1 \\ Ay_1 \\ Az_1 \end{bmatrix} = \begin{bmatrix} 1 + S_x & M_{xy} & M_{xz} \\ M_{yx} & 1 + S_y & M_{yz} \\ M_{zx} & M_{zy} & 1 + S_z \end{bmatrix}^{-1} \begin{bmatrix} A_x - b_x \\ A_y - b_y \\ A_z - b_z \end{bmatrix} \quad (1)$$

where,

$[Ax_1 \ Ay_1 \ Az_1]$ represents calibrated accelerometer data $M_{xy}, M_{xz}, M_{yx}, M_{yz}, M_{zx}, M_{zy}$ indicates Misalignment errors, $[S_x \ S_y \ S_z]$ represents scale factor, $[Ax \ Ay \ Az]$ denotes accelerometer raw data, $[b_x \ b_y \ b_z]$ represents accelerometer bias factor.

Equation (1) can be written as,

$$\begin{bmatrix} Ax_1 \\ Ay_1 \\ Az_1 \end{bmatrix} = \begin{bmatrix} Ac_{11} & Ac_{12} & Ac_{13} \\ Ac_{21} & Ac_{22} & Ac_{23} \\ Ac_{31} & Ac_{32} & Ac_{33} \end{bmatrix} \times \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} + \begin{bmatrix} Ac_{10} \\ Ac_{20} \\ Ac_{30} \end{bmatrix} \quad (2)$$

Scale factor, bias factor and misalignment can be found by using least Square Method as follows

$$Y = W * X \quad (3)$$

where, X denotes calibration parameters that need to be determined, W represents accelerometer raw data collected at 108 stationary positions, Y is Normalized Earth gravity vector.

By combining measurements from all 108 position, X can be solved as follows to determine the calibration parameters.

$$X = [W^T W]^{-1} * W^T * Y \quad (4)$$

Where X will be

$$X = \begin{bmatrix} Ac_{11} & Ac_{12} & Ac_{13} \\ Ac_{21} & Ac_{22} & Ac_{23} \\ Ac_{31} & Ac_{32} & Ac_{33} \\ Ac_{10} & Ac_{20} & Ac_{30} \end{bmatrix}$$

Now this X matrix contains all the three error parameters like scale factor, bias factor and misalignment error.

where,

$[Ac_{10} \ Ac_{20} \ Ac_{30}]$ = Bias factor

$[Ac_{11} \ Ac_{22} \ Ac_{33}]$ = scale factor

$\begin{bmatrix} 1 & Ac_{12} & Ac_{13} \\ Ac_{21} & 1 & Ac_{23} \\ Ac_{31} & Ac_{32} & 1 \end{bmatrix}$ = misalignment error

2) Gyroscope calibration

The Error Model of Tri-axial Gyroscope can be represented by

$$\begin{bmatrix} Gx_1 \\ Gy_1 \\ Gz_1 \end{bmatrix} = \begin{bmatrix} 1 + S_{gx} & N_{xy} & N_{xz} \\ N_{yx} & 1 + S_{gy} & N_{yz} \\ N_{zx} & N_{zy} & 1 + S_{gz} \end{bmatrix}^{-1} \begin{bmatrix} G_x - b_{gx} \\ G_y - b_{gy} \\ G_z - b_{gz} \end{bmatrix} \quad (5)$$

where,

$[Gx_1 \ Gy_1 \ Gz_1]$ = calibrated Gyroscope data

$N_{xy}, N_{xz}, N_{yx}, N_{yz}, N_{zx}, N_{zy}$ = non-orthogonality

$[S_{gx} \ S_{gy} \ S_{gz}]$ = gyroscope scale factor

$[G_x \ G_y \ G_z]$ = gyroscope raw data

$[b_{gx} \ b_{gy} \ b_{gz}]$ = gyroscope bias factor

Equation (5) can be written as,

$$\begin{bmatrix} Gx_1 \\ Gy_1 \\ Gz_1 \end{bmatrix} = \begin{bmatrix} Gc_{11} & Gc_{12} & Gc_{13} \\ Gc_{21} & Gc_{22} & Gc_{23} \\ Gc_{31} & Gc_{32} & Gc_{33} \end{bmatrix} \times \begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix} + \begin{bmatrix} Gc_{10} \\ Gc_{20} \\ Gc_{30} \end{bmatrix} \quad (6)$$

$$V = L * K \quad (7)$$

Scale factor, bias factor and non-orthogonality error can be found by using least Square Method. Where, K refers to calibration parameters that need to be determined, L indicates Gyroscope raw data collected at different rotational speed and V is Normalized angular rate vector. In order to accurately measure the speed of the system, we can consider the rotation of the earth, which can serve as a reference for calibration of the system. However, as the rotation rate has minimal impact on the data, it can be neglected.

The calibration parameter can be computed by integrating all the measurements from 150 dps to 900 dps.

Accordingly, K can be solved as,

$$K = [L^T L]^{-1} * L^T * M \quad (8)$$

Where K will be

$$K = \begin{bmatrix} Gc_{11} & Gc_{12} & Gc_{13} \\ Gc_{21} & Gc_{22} & Gc_{23} \\ Gc_{31} & Gc_{32} & Gc_{33} \\ Gc_{10} & Gc_{20} & Gc_{30} \end{bmatrix}$$

Now this K matrix contains all the three error parameters like scale factor, bias factor and non-orthogonality.

Where,

$[Gc_{10} \ Gc_{20} \ Gc_{30}]$ = Bias factor

$[Gc_{11} \ Gc_{22} \ Gc_{33}]$ = scale factor

$\begin{bmatrix} 1 & Gc_{12} & Gc_{13} \\ Gc_{21} & 1 & Gc_{23} \\ Gc_{31} & Gc_{32} & 1 \end{bmatrix}$ = non-orthogonality

3) Magnetometer calibration

Quantitative data is collected from several position and aggregated to determine the unknown variables (offsets, gains, and cross-axis gains).

The calibration error parameters for magnetometer are calculated by using ellipsoidal fitting algorithm. The basic equation of an ellipsoidal in three-dimension is

$$aX^2 + bY^2 + cZ^2 + d2XY + e2XZ + f2YZ + g2X + h2Y + i2Z = 1 \quad (9)$$

The data matrix can be constructed as,

$$D = [X^2, Y^2, Z^2, 2XY, 2XZ, 2YZ, 2X, 2Y, 2Z]$$

It is possible to calculate the least-square error approximation for the unknowns in L by utilizing the pseudo-inverse of the non-square matrix.

$$L = (D^T D)^{-1} (D^T 1) \quad (10)$$

After solving this we will get the solution as,

$$L = [a, b, c, d, e, f, g, h, i]$$

Offset:

$$\begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix} * \begin{bmatrix} O_x \\ O_y \\ O_z \end{bmatrix} = - \begin{bmatrix} g \\ h \\ i \end{bmatrix} \quad (11)$$

After obtaining the offsets, a further auxiliary matrix B_4 is calculated, which represent the ellipsoid adjusted to the origin.

$$B_4 = E * F * E^T$$

Where,

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ O_x & O_y & O_z & 1 \end{bmatrix}$$

$$F = \begin{bmatrix} a & d & e & g \\ d & b & f & h \\ e & f & c & i \\ g & h & i & -1 \end{bmatrix}$$

$$B_3 = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ -b_{44} \end{bmatrix}$$

Gains and cross-axis gains can be computed from eigenvalues and eigenvectors, respectively, of B_3 .

$$G = \sqrt{\frac{1}{\text{diag}(\text{eigen value}(B_3))}} \quad (12)$$

Where,

G = gain

$$R = [\text{eigenvector}(B_3)] \quad (13)$$

Where,

R = cross-axis gain.

B. Dynamic calibration

The static calibration only provides a single calibration result and is only effective for linear systems. If we work on real-time data processing, then we require field calibration, which often exhibits non-linear behaviour. Static calibration does not provide accurate results in this case, as it does not account for temperature changes, vibrations, and aging. To achieve proper accuracy in dynamic model calibration, we need to utilize alternative methods such as the Unscented Kalman filter. The initial estimations for the scale factor, bias factor, and misalignment error are determined based on the results obtained during static calibration. The state vector x includes the calibration parameters for the accelerometer, gyroscope, and magnetometer.

$x = [b_{acc}, s_{acc}, m_{acc}, b_{gyr}, s_{gyr}, m_{gyr}, b_{mag}, s_{mag}, m_{mag}]$
 b_{acc} , b_{gyr} are Accelerometer and gyroscope bias factor, s_{acc} , s_{gyr} are accelerometer and gyroscope scale factor, m_{acc} is Accelerometer misalignment error, m_{gyr} is Gyroscope non-orthogonality error, b_{mag} is Magnetometer offset, s_{mag} is magnetometer gain, m_{mag} is Magnetometer Cross-gain (3×3).

1. Unscented Kalman Filter

In contrast to the linearization approach of the EKF, the UKF employs a deterministic sampling technique known as the unscented transform to provide a more precise

approximation of the mean and covariance of the state distribution. The transformation process entails producing a collection of sigma points that accurately represent the mean and variance of the state distribution. Static calibration provides a one-time calibration. In contrast, the UKF allows for continuous, real-time updates of calibration parameters as new data becomes available.

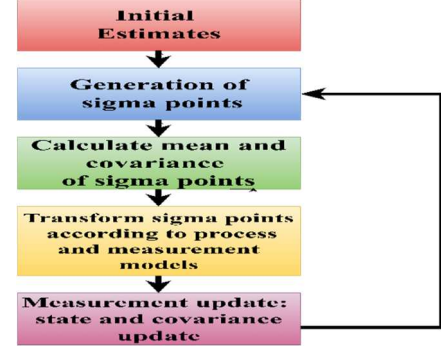


Fig. 1: Flow chart of UKF

Figure 1 explains about the unscented Kalman filter algorithm which is designed based on [12]. Here the UKF is implemented only to get accurate calibration result when we are going to perform dynamic model. Figure 2 depicts the dynamic calibration for estimating the calibration

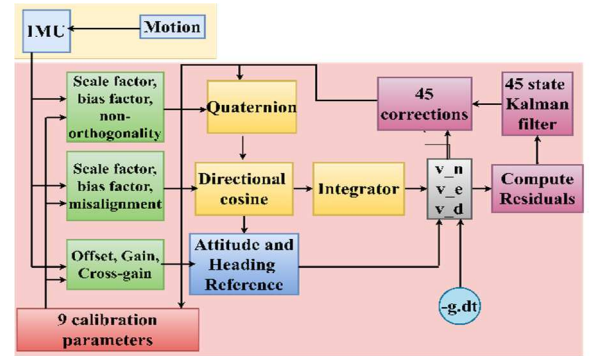


Fig. 2: Flow chart of dynamic calibration

parameters. UKF requires initial estimates, which must be accurate, so the initial estimates obtained from the static calibration parameters are fed as input to UKF to get more precise output. After the first iteration the estimated the Kalman filter output (corrected state estimates) is fed back into the computation process. This loop ensures that the algorithm is constantly updating and refining its estimates.

III. EXPERIMENTAL SETUP AND CALIBRATION PROCEDURE

A) Accelerometer

The 9-axis ICM20948 IMU module is used for the experiment. The developed calibration setup consists of Aluminium plate base which is attached to a wooden table and rotatable through screw arrangement, a 360° Pro circle (Protractor) to measure the rotation angle of the plate, Bluetooth module for wireless data transfer, micro-controller and Battery. The IMU module is mounted on the plate and rotated at different angle for all 3-axis. Firstly, the IMU is placed along Z-axis and the setup is rotated whole 360° at 10-degree intervals. This entails rotating at intervals

of 10 degrees and maintaining that position for 30 seconds, while recording the acceleration data in all three axes. Likewise, the same procedure is to be performed for 360-degree rotation to obtain 36 sets of positional data. Likewise, IMU module is detached from the aluminum plate and subsequently reposition it on the aluminum plate with the Y-axis facing upwards. Again, the process is to be repeated for 360 degrees in order to obtain 36 different positions. Likewise, the IMU is being detached and then reposition it on the plate with the X-axis oriented upwards.

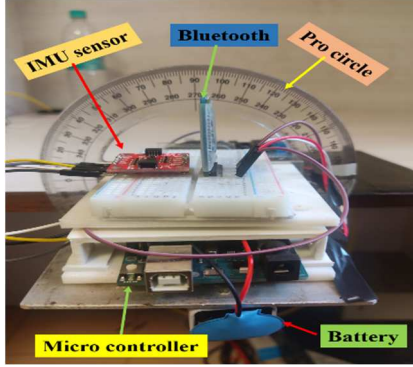


Fig. 3: Accelerometer calibration setup

The process is repeated for a full 360 degrees in order to obtain 36 different positions once again. Hence, in total, there will be data for 108 positions. Subsequently, the vector is created that represents the force of gravity acting on the Earth. This refers to identify the axis with the highest absolute value in the acceleration data of all three axes. If the highest value is a positive integer, assign 1 to that axis. If it is a negative integer, assign -1 to that axis. IMU raw data (W) and the normalized Earth gravity vector (Y) are constructed in this manner.

B) Gyroscope

The gyroscope calibration setup comprises a system for controlling speed. The IMU sensor is positioned on the spin table with the Z-axis facing upwards. The gyroscope is activated and remain idle until the cold start time has elapsed. 100 gyro data samples are collected and moving average filter is used to get the zero-rate level (R_{x0} , R_{y0} , R_{z0}). When the gyroscope is stationary for a few minutes, the zero-rate-in run bias is determined and accounted for any minor temperature fluctuations. The spin table is rotated in clockwise at a rate of 150 degrees per second and waited for the angular velocity of the spin table to stabilize. Subsequently, the data is gathered for about 3 minutes. Continuously the spinning table is rotated at an interval of 150 degrees per second until it reaches a speed of 900 degrees per second.

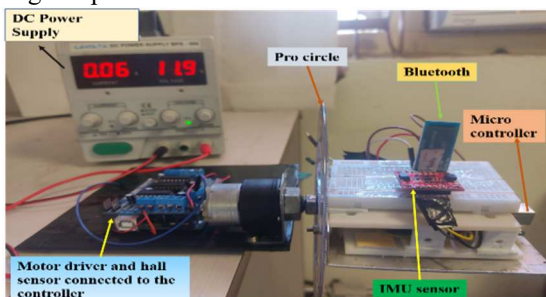


Fig. 4: gyroscope calibration setup

Similarly, the spin table is rotated in an anticlockwise direction, ranging from 150 degrees per second (dps) to 900 dps, with an interval of 30 dps. The data are recorded during this rotation. Next, IMU is placed on the spin table with the X-axis facing upwards. The same procedure is reiterated for both clockwise and counter-clockwise orientation. Again, IMU is placed on the spin table with the Y-axis facing upwards and rotated both clockwise and counter-clockwise directions, ranging from 150 degrees per second to 900 degrees per second, with an interval of 30 degrees per second. Accordingly, the known applied angular rate data matrix and the gyroscope's raw data matrix are constructed. Then by applying the Least Squares Fit algorithm, the calibration parameters are determined.

C) Magnetometer

To calibrate the magnetometer, rotate the IMU in all directions to capture the data, which will contain the strength of the magnetic field in each axis. To measure only the earth magnetic field with a magnetometer, all other time-varying magnetic anomalies must be eliminated. The actual stimulus is then measured as the modulus of the earth's magnetic field.

IV. EXPERIMENTAL RESULT AND DISCUSSION

After determining the static calibration data from 108 positions for accelerometer, we got scale factor, bias factor, and misalignment error by using the tri-axial sensor error model. After solving equation (3), we will get the scale factor, bias factor and misalignment error now these factors are added to equation (1) to obtain the calibrated data. The calibrated and uncalibrated data for some positions are shown in Table I.

TABLE I. COMPARISON OF ACCELEROMETER CALIBRATED VS UNCALIBRATED DATA

	X-axis output		Y-axis output		Z-axis output	
	Uncalibrated	calibrated	Uncalibrated	Calibrated	Uncalibrated	Calibrated
X-axis UP	1.460	0.979	-0.380	-0.051	-0.390	0.074
X-axis Down	-1.380	-1.022	-0.320	-0.062	-0.190	-0.013
Y-axis UP	0.280	-0.025	0.955	0.971	0.020	0.053
Y-axis Down	0.350	0.025	-1.011	-0.983	-0.247	0.045
Z-axis UP	-0.167	-0.065	-0.128	-0.07	1.107	1.003
Z-axis Down	0.341	0.029	-0.951	-0.143	-0.934	-0.971

One example is shown by facing the z-axis of IMU upward at 0 degree. The data are plotted for calibrated and uncalibrated data. Figure 5, 6 and 7 explain the triaxial acceleration readings acquired before and after calibration.

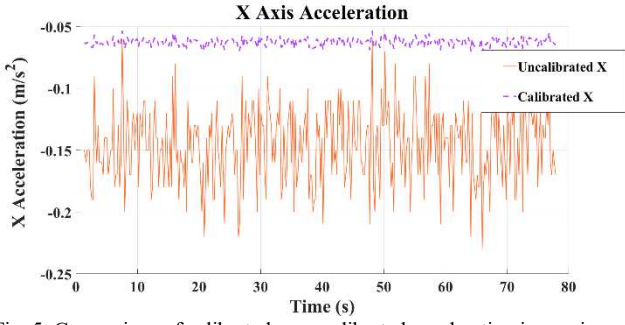


Fig. 5: Comparison of calibrated vs uncalibrated acceleration in x-axis

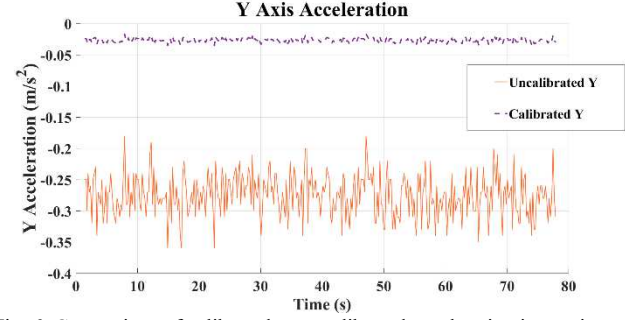


Fig. 6: Comparison of calibrated vs uncalibrated acceleration in y-axis

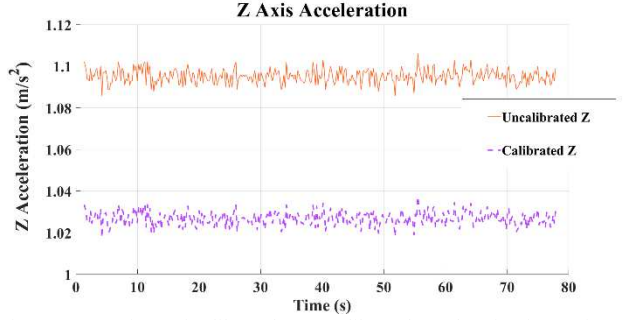


Fig. 7: Comparison of calibrated vs uncalibrated acceleration in z-axis

TABLE II. ACCELEROMETER CALIBRATION PARAMETERS

	Coefficient	Static method	UKF
Bias	Ac_{10}	0.0326	0.0531
	Ac_{20}	-0.0210	-0.0537
	Ac_{30}	-0.1765	-0.2368
Scale Factor	Ac_{11}	1.0031	0.9926
	Ac_{22}	0.9977	1.0143
	Ac_{33}	1.0112	0.9857
Misalignment	Ac_{12}	0.0064	0.0048
	Ac_{21}	-0.0067	-0.0074
	Ac_{23}	0.0106	0.0095
	Ac_{32}	0.0213	0.0278
	Ac_{13}	0.0075	0.0067
	Ac_{31}	0.0062	0.0054

Table II indicates the calibration parameters, like scale factor, bias factor, and misalignment, which are obtained by collecting data from 108 points and integrating it into the least square fit algorithm. The estimated states from UKF are also shown in last column, which are required for dynamic calibration.

Next, the magnetometer calibration is performed. To collect data from the magnetometer, the IMU is manually rolled constantly in three dimensions. The captured data are applied to the ellipsoid fitting algorithm. Here around 3000 samples of data are collected and iterated. The offset and

gain are obtained using the ellipsoid fitting method as follows,

$$p_{corrected} = (p - O) * A \quad (14)$$

Here p is a 1×3 matrix of uncalibrated magnetometer measurements, O is the offset obtained from magnetometer calibration, and $p_{corrected}$ is the 1×3 matrix of corrected magnetometer measurements, which lie on a sphere.

Now using equation (14), the uncalibrated or raw magnetometer data are corrected. Figure 7 and 8 illustrate the comparison between the uncalibrated and calibrated data.

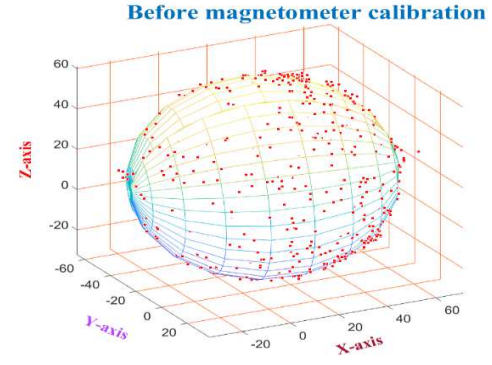


Fig. 8: Ellipsoidal fit

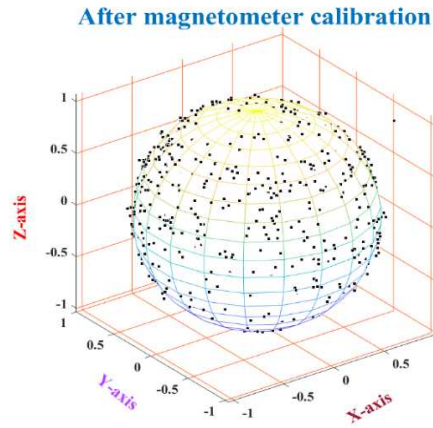


Fig. 9: Normalized to unit sphere

Subsequently, the gyroscope calibration is done. The gyroscope calibration is entirely different than the accelerometer calibration. Although the algorithm will be similar but the data captured for the calibration process is different. Here by using a motor, we are rotating the entire setup from 150 dps to 900 dps with 30 dps interval at constant speed. While the system is in rotation, it is not possible to physically link the microcontroller to the computer. Therefore, bluetooth is used to transfer the data wirelessly. As a result, the rotation platform can be rotated at a uniform speed according to our predetermined setting.

TABLE III. GYROSCOPE CALIBRATION PARAMETERS

	coefficient	Static method	UKF
Bias	Gc_{10}	0.1834	0.1683
	Gc_{20}	-0.6944	-0.6352
	Gc_{30}	-0.9476	-0.9881

Scale Factor	G_{c11}	1.0472	1.0265
	G_{c22}	1.1637	1.1749
	G_{c33}	0.9836	0.9684
Non-orthogonality	G_{c12}	0.0176	0.0148
	G_{c21}	-0.0285	-0.0298
	G_{c23}	0.0098	0.0175
	G_{c32}	-0.0041	-0.0036
	G_{c13}	0.0125	0.0097
	G_{c31}	-0.0081	-0.0126

Table III indicates the calibration parameters, like scale factor, bias factor, and non-orthogonality, that were acquired by capturing the data from 150 dps to 900 dps with 30 dps interval at constant speed and integrating it into the least square fit algorithm. One example is shown by facing the z-axis of IMU towards upward and data are plotted for calibrated and uncalibrated values. Figure 3,4 and 5 explain the triaxial angular velocity readings acquired before and after calibration.

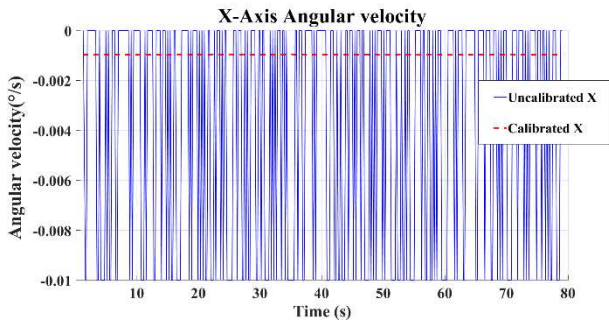


Fig. 10: Comparison of calibrated vs uncalibrated angular velocity in x-axis

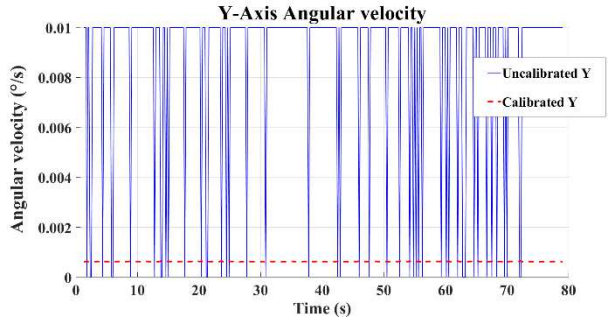


Fig. 11: Comparison of calibrated vs uncalibrated angular velocity in y-axis

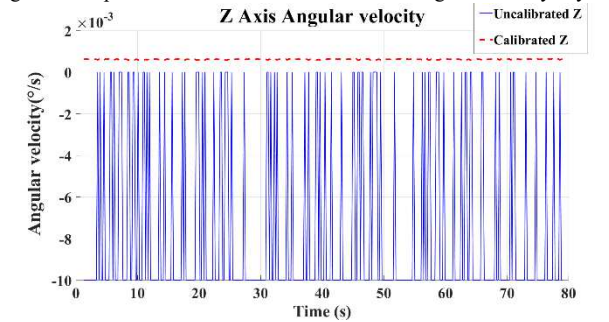


Fig. 12: Comparison of calibrated vs uncalibrated angular velocity in z-axis

V. CONCLUSION

An innovative calibration method has been developed for 9-axis IMU comprising of accelerometer, gyroscope, and magnetometer. A cost-effective calibration setup has been devised to quickly calibrate the sensors. By implementing

the suggested approach, it is possible to calibrate any IMU in-field with a few standard rotations within around 15 minutes. The approach uses solely the gravitational field of the Earth as a reference for calibration, therefore removing any limitations on the orientation of the sensor during the calibration process. By using UKF in dynamic calibration the precision is increased.

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